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Proposition[section]

On the Construction of Some Lychrel Numbers

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Abstract

For decades now, Lychrel numbers have been a passionate topic for many researchers. Their construction leads to the construction of palindromes which are important in biotechnology, notably in DNA and RNA sequences, as they help to stabilize these types of molecules (5). In some basis, the existence of these numbers has been proven. It is the case in base **2**. In others, just candidates are known. It is the case in base **10** where a list of Lychrel candidates (smaller than **10000**) has been computed. The question answered in this piece of paper is presented as follow: is there a non trivial relation between Lychrel numbers or Lychrel candidates such that some of them can be obtained from others via these relations? We answer this question by proposing in bases between **3** and **10** some mathematical approaches of extracting new Lychrel numbers or new Lychrel candidates from others, and by applying these approaches on some cases, notably in base **4** and base **10**

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1 Introduction

In Numbers Theory, the **reverse number** of a natural number $a = a_1...a_n$ in base b (where $n = length(a)$ designates the number of digits of a), which we denote by $Rev(a)$ is the natural number such that $Rev(a) = a_n a_{n-1}...a_1$. A **palindrome** number is a natural number such that $a = Rev(a)$. For example, all natural numbers in base 10 of the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ are palindromes. It is also the case with the following set of natural numbers

$\{11, 22, 33, 44, 55, 66, 77, 88, 99\}$. If a natural number a is not a palindrome, one can eventually obtain a palindrome by iteratively adding a to its reverse number. For example, $124 + Rev(124) = 124 + 421 = 545$ which is a palindrome. With $a = 159$, we have $159 + Rev(159) = 159 + 951 = 1110$, and $1110 + Rev(1110) = 1110 + 0111 = 1221$. Hence, we obtained a palindrome after two iterations. The natural number 89 leads to a palindrome after 24 iterations. However, they are natural numbers that do not seem to form a palindrome after the computation of the process of reversing and adding their reverse numbers. These numbers have become known as **Lychrel Numbers**. In base 2, it as been shown that 22 is a Lychrel number. In base 10, no Lychrel number has yet been found, even if some candidates do exist, notably $196, 879, 1997...$ A more detailed definition of Palindromes and Lychrel numbers can be found in (3).

However, until today, all Lychrel numbers or Lychrel candidates are still obtained after the iterative process of reversal and addition of the reverse number is tested. Hence, this test itself remains a difficult task, especially when the number of digits becomes large and the threshold very hight. In this situation, the computer program may run for days, weeks, months and even years before producing the result. In (9), with the number 196 in base 10, the computer program of the iterative process of reversing and adding the reverse number ran for three years before reaching the threshold without producing a palindrome. We then ask the question whether some Lychrel numbers or Lychrel candidates can be obtained from others without the iterative process of reversal and addition of the reverse number been tested.

To the best of our knowledge, the problem of not reaching a palindrome after taking a natural number and iteratively reversing it and adding its reverse number was first raised in (6) when the author mentions the number of natural numbers for which no palindrome is reached after 100 successive reversal and addition of their reverse numbers. Since then, the existence of Lychrel numbers has been proven in many bases, among which the bases 2, 11, 17 or 26, and presented in (3). In base 10, some works has been done to verify whether the natural number 196 is a Lychrel candidate: in 1987, The author from (9) wrote an algorithm to verify if the natural number 196 is a Lychrel number, and this algorithm stop running three years later after reaching a natural non palindrome number with 1 million digits and after approaching $2,416$ millions iterations. In (4), the research continued with a super computer and in 1995, the author obtained a natural number with 2 millions digits, which is not a palindrome. Jason Doucette took-over the research in the year 2000 and obtained a natural number with $12,5$ millions digits, which is still not a palindrome. In 2006, Wade VanLandingham continued with the computation and reached a natural number with 300 millions digits and which is also not a palindrome. Romain Dolbeau took over the work and after more than $2,4$ billion iterations, he obtained in 2015 a non palindrome number with a billion digits. The summary of works done on 196 and other Lychrel numbers can be found in (7) and (8). In 2017, the authors from (1) established that the

existence of Lychrel numbers in base 10 is due to the symmetry breaking in the space of the natural numbers. However, their demonstration remains a global one because they was not able to take a particular natural number and prove that it is a Lychrel number.

The rest of the work is organized as follows: In section 2 we explain how to obtain the length and the expression of an iteration. In section 3 we present some properties to extract new Lychrel numbers or Lychrel candidates from others and provide some examples. In section 4 we apply the defined properties to some cases in base 4 and base 10. In section 5 we focus on testing some new numbers obtained in base 10. Section 6 addresses the new notion of Palindrome-Lychrel numbers or Palindrome-Lychrel candidates and section 7 concludes the paper.

2 Length and Expression of an iteration

In this section, we define the notion of iteration and provide some tools which help us to explain how to obtain the length and the expression of an iteration.

Let $a = a_1...a_n$ be a natural number in base b ($3 \leq b \leq 10$), where for all $i \in \{1, ..., n\}$, a_i is a digit of a. The length of a in this work, denoted by $l(a)$, designates the number of digits of a. Moreover, considering a natural number $a = a_1...a_n$ and its reverse number $Rev(a) = a_n...a_1$, an **iteration** is the operation which consists of adding a to $Rev(a)$. Hence, we define the **unity function** $\mathcal{U}$ as followed

$$\begin{aligned} \mathcal{U} : N &\longrightarrow N \\ a &\longmapsto a_{l(a)} \end{aligned}$$

Some examples in base 10 are contained in the following table,

a	0	3	25	1268	1000	0001	7960	1997	40961	20000	178997
$l(a)$	1	1	2	4	4	4	4	4	5	5	6
$\mathcal{U}$(a)	0	3	5	8	0	1	0	7	1	0	7

The unity function $\mathcal{U}$ verifies some elementary properties:

Lemma 2.1 *let $a \neq 0$ be a digit of a natural number in base b ($3 \leq b \leq 10$), and $\rho \in \{0, 1\}$.*

1i) For all $x, y \in N$, $\mathcal{U}(x + y) = \mathcal{U}(\mathcal{U}(x) + \mathcal{U}(y))$;
2i) For all $x \in N$, if $\mathcal{U}(x) \in \{0, ..., b - 2\}$ then $\mathcal{U}(x + \rho) = \mathcal{U}(x) + \rho$;
3i) For all $i \in \{1, ..., 10\}$, if $a + i \geq 10$, then $\mathcal{U}(a + i) = a - 10 + i$.

Proof 1i) Let x and y be two natural numbers. Then x and y are respectively expressed in base b as $x = \sum_{i=0}^{l(x)-1} x_i.b^i$, $y = \sum_{j=0}^{l(y)-1} y_j.b^j$. Moreover the

last term of $x+y$ is x_0+y_0, which implies that $\mathcal{U}(x+y)=\mathcal{U}(x_0+y_0)$. Hence, $\mathcal{U}(x) = x_0$ and $\mathcal{U}(y) = y_0$ and then $\mathcal{U}(\mathcal{U}(x)+\mathcal{U}(y))=\mathcal{U}(x_0 + y_0)= \mathcal{U}(x + y)$.

2i) Let x be a natural number such that $\mathcal{U}(x) \in \{0, ..., b-2\}$. Then $0 \leq \mathcal{U}(x) + \rho \leq b-1$ because $\rho \in \{0, 1\}$. Hence, $\mathcal{U}(\mathcal{U}(x)+\rho)=\mathcal{U}(x)+\rho$. Also, $\mathcal{U}(\mathcal{U}(x)+\rho)=\mathcal{U}(\mathcal{U}(x)+\mathcal{U}(\rho))$ (since $\rho \in \{0, 1\}$), and $\mathcal{U}(\mathcal{U}(x)+\mathcal{U}(\rho))=\mathcal{U}(x+\rho)$ by $1i$). It then comes that $\mathcal{U}(x + \rho)=\mathcal{U}(x) + \rho$.

3i) Let i be a natural number such that $i \in \{1, ..., 10\}$ and $a + i \geq 10$. Then $20 - 1 \geq a + i \geq 10$, since a is a digit in base b and $a + i \geq 10$. Hence, $10 - 1 \geq a - 10 + i \geq 0$ and then $\mathcal{U}(a - 10 + i) = a - 10 + i$. We then have $\mathcal{U}(a + i)=\mathcal{U}((a - 10 + i) + 10)=\mathcal{U}(\mathcal{U}(a - 10 + i)+\mathcal{U}(10))$ (because of $1i$)) $=\mathcal{U}(\mathcal{U}(a - 10 + i)+0)=\mathcal{U}(a - 10 + i) = a - 10 + i$.

□

Hence, in base b ($3 \leq b \leq 10$), if $a \neq 0$ is a digit of a natural number, then $\mathcal{U}(a + 10) = a$ and $\mathcal{U}(a + 10 - 1) = a - 1$.

Also, we consider the **iteration function** ϕ, which is actually the one defined in (1), as follow:

$$\begin{aligned} \phi : N &\longrightarrow N \\ a &\longmapsto a + Rev(a) \end{aligned}$$

Then, the first iteration of a will be denoted by $\phi^1(a) = \phi(a)$, the second iteration will be $\phi^2(a) = \phi(\phi^1(a)) = \phi(\phi(a))$ and for all $n \in N$, $\phi^n(a) = \phi(....(\phi(a)))$ (n factors). For example, in base 10, if $a = 75$, then $\phi^2(75) = \phi(\phi(75)) = \phi(75) + Rev(\phi(75)) = 75 + Rev(75) + Rev(75 + Rev(75)) = 75 + 57 + Rev(75 + 57) = 132 + Rev(132) = 132 + 231 = 363$.

Let $a = a_1...a_n$ be a natural number in base b, where $3 \leq b \leq 10$. Let set $\phi(a) = a + Rev(a)$. Then the nth iteration of a can be expressed as a function of a as $\phi^n(a) = a + \sum_{i=1}^{n-1} Rev(\phi^i(a))$. As for the first iteration, its length is given by $l(\phi(a)) = l(a) + \alpha$, where $\alpha \in \{0, 1\}$. $\alpha = 1$ if and only if $a_1 + a_{l(a)} \geq 10$ or there exists $i \in \{1, ..., l(a)\}$ such that $a_i + a_{l(a)-i+1} \geq 10$ and $a_j + a_{l(a)-j+1} = b - 1$ for all $j \leq i - 1$. For example, in base 10, if $a = 4006$, then we have $l(a) = 4$ and $a_1 + a_{l(a)} = 4 + 6 = 10 \geq 10$. Hence, $\alpha = 1$ and then $l(\phi(a)) = l(a) + 1 = 4 + 1 = 5$. In the second example, also in base 10, we consider the case where $a = 456745$. Then for $j = 3$, $a_3 + a_4 = 6 + 7 = 13 \geq 10$ and $a_j + a_{7-j} = 9$ for all $j \leq 2$. Hence, $\alpha = 1$ and then $l(\phi(a)) = l(a) + 1 = 6 + 1 = 7$.

In order to successfully add a to its reverse number to obtained the first iteration of a, we define the parameter $\epsilon(a)$ which depend both of the length and the digits of a, as follows:

$$\epsilon_j(a) = \begin{cases} 0 \; if & j = l(a) + \alpha \\ 1 \; if & j < l(a) + \alpha \;\; and \;\; a_j + a_{l(a)-j+1} + \epsilon_{j+1}(a) \geq 10 \\ 0 & else \end{cases}$$

It is important to note that The $\epsilon_j(a)$ are calculated in the decreasing order. For all $j \in \{1, ..., l(a) + \alpha\}$, one can also assimilate $\epsilon_j(a)$ to the function $1_{\{a_j + a_{l(a)-j+1} + \epsilon_{j+1}(a) \geq 10\}}$ which is equal to 1 if $a_j + a_{l(a)-j+1} + \epsilon_{j+1}(a) \geq 10$, and equal to 0 otherwise.

Then the following are the expressions, depending of α, of the first iteration of the natural number a.

If $\alpha = 1$, then:

- $l(\phi(a)) = l(a) + 1$,
- $\phi(a)_1 = 1$,
- $\phi(a)_{l(\phi(a))} = \mathcal{U}(a_{l(a)} + a_1)$,
- For every $j \in \{2, ..., l(a) + 1\}$, $\phi(a)_j = \mathcal{U}(a_{j-1} + a_{l(a)-j+2} + \epsilon_j(a))$.

If $\alpha = 0$, then:

a) $l(\phi(a)) = l(a)$,
b) $\phi(a)_1 = \mathcal{U}(a_1 + a_{l(a)} + \epsilon_1(a))$,
c) $\phi(a)_{l(\phi(a))} = \mathcal{U}(a_{l(a)} + a_1)$,
d) For every $j \in \{2, ..., l(a)\}$, $\phi(a)_j = \mathcal{U}(a_j + a_{l(a)-j+1} + \epsilon_j(a))$.

For example, in base 10, we consider the case where $a = 974$, we have $l(a) = 3$, $a_1 + a_{l(a)} = a_1 + a_3 = 9 + 4 = 13 \geq 10$. Therefore $l(\phi(a)) = l(a) + 1 = 3 + 1 = 4$. Moreover, we determine the epsilons in the decreasing order as follows: $\epsilon_4(a) = 0$, $\epsilon_3(a) = 1$ because $a_3 + a_1 \geq 10$, $\epsilon_2(a) = 1$ because $a_2 + a_2 + \epsilon_3(a) \geq 10$ and $\epsilon_1(a) = 1$ because $a_1 + a_3 + \epsilon_2(a) \geq 10$. Hence, we can determine the digits of $\phi(a)$ as follows:

i) $\phi(a)_1 = 1$,
ii) $\phi(a)_{l(\phi(a))} = \phi(a)_4 = \mathcal{U}(a_{l(a)} + a_1) = \mathcal{U}(a_3 + a_1) = \mathcal{U}(4 + 9) = \mathcal{U}(13) = 3$,
iii) $\phi(a)_3 = \mathcal{U}(a_2 + a_{l(a)-1} + \epsilon_3(a)) = \mathcal{U}(7 + 7 + \epsilon_3(a)) = \mathcal{U}(14 + \epsilon_3(a)) = \mathcal{U}(14 + 1) = \mathcal{U}(15) = 5$,
iv) $\phi(a)_2 = \mathcal{U}(a_1 + a_{l(a)} + \epsilon_2(a)) = \mathcal{U}(9 + 4 + \epsilon_2(a)) = \mathcal{U}(13 + \epsilon_2(a)) = \mathcal{U}(13 + 1) = \mathcal{U}(14) = 4$.

Hence, $\phi(a) = \phi(a)_1\phi(a)_2\phi(a)_3\phi(a)_4 = 1453$.

In the more general way, if a and m are two natural numbers in base b ($3 \leq b \leq 10$) with $l(a)$ and $l(m)$ the respective lengths of a and m. Let say that our aim is to make the addition of the two numbers in base b. Then the length of $a + m$ is given by $\max(l(a), l(m)) + \alpha$ (with $\alpha \in \{0, 1\}$). $\alpha = 1$ if and only if $a_1 + b_1 \geq 10$ or there exists $i \in \{1, ..., \max(l(a), l(m))\}$ such that $a_i + m_i \geq 10$ and $a_j + m_j = b - 1$ for all $j \leq i - 1$. For example, in base 10, if $a = 900$ and $m = 105$, then we have $l(a) = l(m) = 3$ and $a_1 + m_1 = 9 + 8 = 17 \geq 10$. Hence, $\alpha = 1$ and then $l(a + m) = \max(l(a), l(m)) + 1 = l(a) + 1 = 3 + 1 = 4$. The second example also in base 10 presents the case where $a = 4545$ and $m = 5456$, then for $j = 4$, $a_4 + m_4 = 11 \geq 10$ and $a_j + m_j = 9$ for all $j \leq 3$. Hence, $\alpha = 1$ and then $l(a + m) = \max(l(a), l(m)) + 1 = l(a) + 1 = 4 + 1 = 5$.

In the above examples, we have $l(a) = l(m)$. If $l(a) \neq l(m)$, then we make the following adjustments: if $l(m) < l(a)$, then 0 is added $l(a) - l(m)$ times to

m and before the digit m_1. For example, in base 10, if $a = 105$ and $m = 9$, then $l(a) - l(m) = 3 - 1 = 2$. Hence 0 is added 2 times to m and before 9. m then becomes $d = 009$ with a greater length $l(d) = 3 = l(a)$ and $a+m = a+d$. The same situation occurs with a if $l(a) < l(m)$.

We then define the parameter $\sigma(a, m)$ which depends both of the digits of a and m as follows:

$$\sigma_j(a,b) = \begin{cases} 0 \; if & j = \max(l(a), l(m)) + \alpha \\ 1 \; if & j < \max(l(a), l(m)) + \alpha \;\; and \;\; a_j + m_j + \sigma_{j+1}(a,m) \geq 10 \\ 0 & else \end{cases}$$

The $\sigma_j(a, m)$ are also calculated in a decreasing order. For all $j \in \{1, ..., \max(l(a), l(m)) + \alpha - 1\}$, one can also assimilate $\sigma_j(a, m)$ to the function $1_{\{a_j+m_j+\sigma_{j+1}(a,m)\geq 10\}}$ which is equal to 1 if $a_j + m_j + \sigma_{j+1}(a, m) \geq 10$, and 0 otherwise.

The digits of $a + m$ are then obtained as follows:

If $\alpha = 1$, then:

- $l(a + m) = l(a + d) = l(d) + 1$,
- $(a + m)_1 = (a + d)_1 = 1$,
- $(a + m)_{l(a+d)} = (a + d)_{l(a+d)} = \mathcal{U}(a_{l(a)} + d_{l(d)})$,
- For every $j \in \{2, ..., l(d)+1\}$, $(a+m)_j = (a+d)_j = \mathcal{U}(a_{j-1}+d_{j-1}+\sigma_j(a, m))$.

If $\alpha = 0$, then:

a) $l(a + m) = l(a + d) = l(d)$,
b) $(a + m)_1 = \mathcal{U}(a_1 + d_1 + \sigma_1(a, m))$,
c) $(a + m)_{l(a+m)} = (a + d)_{l(a+d)} = \mathcal{U}(a_{l(a)} + d_{l(d)})$,
d) For every $j \in \{2, ..., l(d)\}$, $(a + m)_j = (a + d)_j = \mathcal{U}(a_j + d_j + \sigma_j(a, m))$..

Let us reconsider the two previous numbers, $a = 4545$ and $m = 5456$. We already know that $l(a + m) = 5$. Now we are going to search for the digits of $a + m$. From the definition of σ, we have $\sigma_5(a, m) = 0$. $\sigma_4(a, m) = 1$ because $a_4+m_4 = 5+6 = 11 \geq 10$, $\sigma_3(a, m) = 1$ because $a_3+m_3+\sigma_4(a, m) = 4+5+1 = 10 \geq 10$, $\sigma_2(a, m) = 1$ because $a_2 + m_2 + \sigma_3(a, m) = 5 + 4 + 1 = 10 \geq 10$ and $\sigma_1(a, m) = 1$ because $a_1 + m_1 + \sigma_2(a, m) = 4 + 5 + 1 = 10 \geq 10$. Hence, we can determine the digits of $a+m$ as follow: $(a+m)_1 = 1$, $(a+m)_{l(a+m)} = (a+m)_5 = \mathcal{U}(a_4 + m_4) = \mathcal{U}(5 + 6) = \mathcal{U}(11) = 1$, $(a + m)_4 = \mathcal{U}(a_3 + m_3 + \sigma_4(a, m)) = \mathcal{U}(4 + 5 + 1) = \mathcal{U}(10) = 0$, $(a + m)_3 = \mathcal{U}(a_2 + m_2 + \sigma_3(a, m)) = \mathcal{U}(5 + 4 + 1) = \mathcal{U}(10) = 0$ and $(a + m)_2 = \mathcal{U}(a_1 + m_1 + \sigma_2(a, m)) = \mathcal{U}(4 + 5 + 1) = \mathcal{U}(10) = 0$. Hence, $a + m = 10001$.

In the previous case, $l(a) = l(m)$. Let explore a case where $l(a) \neq l(m)$. We consider the two numbers $a = 4545$ and $m = 56$. Then $l(m) = 2$ and $l(a) = 4$.

We then set $d = 0056$. $a_1+d_1 = 4+0 = 4 < 9$, then $\alpha = 0$ and $l(a+m) = l(a+d) = 4$. Now we are going to search for the digits of $a+m$. From the definition of σ, we have $\sigma_4(a,m) = 0$. $\sigma_3(a,m) = 1$ because $a_4 + m_4 = 5+6 = 11 \geq 10$, $\sigma_2(a,m) = 1$ because $a_3 + m_3 + \sigma_4(a,m) = 4+5+1 = 10 \geq 10$, $\sigma_1(a,m) = 0$ because $a_2 + m_2 + \sigma_3(a,m) = 5+0+1 = 6 < 9$. Hence, we can determine the digits of $a+m$ as follows: $(a+m)_1 = \mathcal{U}(a_1+d_1+\sigma_1(a,m)) = \mathcal{U}(4+0+0) = 4$, $(a+m)_{l(a+m)} = (a+d)_{l(a+d)} = \mathcal{U}(a_{l(a)} + d_{l(d)}) = \mathcal{U}(5+6) = \mathcal{U}(11) = 1$, $(a+m)_3 = \mathcal{U}(a_3+d_3+\sigma_3(a,m)) = \mathcal{U}(4+5+1) = \mathcal{U}(10) = 0$ and $(a+m)_2 = \mathcal{U}(a_2+d_2+\sigma_2(a,m)) = \mathcal{U}(5+0+1) = \mathcal{U}(6) = 6$. Hence, $a+m = 4601$.

3 Some properties of Lychrel numbers

In this section, we present some properties of Lychrel number in some bases. These properties have a double advantage. Firstly, they enable without a necessary use of computer programs, to extract Lychrel numbers from others in bases where this type of number have been found, and secondly they permit without a necessary use of computer programs, to construct new candidates from others in the bases where no Lychrel numbers have yet been detected. The immediate consequence is therefore the drastic reduction of the time used on computers for the search of Lychrel numbers or Lychrel candidates.

Let l be a natural number and N_l the set of all natural numbers of length l in base b ($3 \leq b \leq 10$). Then we define on N_l the binary relation $\mathcal{R}_l$ as follow: $(a,m) \in \mathcal{R}_l$ if and only if for all $i \in \{1,...,l\}$, $a_i + a_{l-i+1} = m_i + m_{l-i+1}$. Then $\mathcal{R}_l$ is reflexive, symmetric and transitive and is then an equivalence relation on the set N_l. In other words, $(a,m) \in \mathcal{R}_l$ if and only if $\phi(a) = \phi(m)$.

The following properties help us to search, for a given number a, the number m verifying $\phi(a) = \phi(m)$ under some conditions.

Proposition 3.1 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If $1 \leq a_1 \leq b-2$ and $a_{l(a)} > 0$, then $\phi(a) = \phi(a + 10^{l(a)-1} - 1)$.*

Proof Let consider such number $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$. We set $x = a + (10^{l(a)-1} - 1)$. Then $l(x) = l(a)$ because $a_1 \leq b-2$ and $l(10^{l(a)-1} - 1) = l(a) - 1$. Moreover, setting $m = 10^{l(a)-1} - 1$, $l(m) = l(a) - 1$. We then consider d the number obtained by adding 0 before the first digit of m. Then $l(a) = l(d)$ and $a + m = a + d$. Hence, $\sigma_{l(x)}(a,d) = 0$ by definition and since in base b, all the digits of d are equal to $b-1$ except the first one and $a_{l(a)} > 0$, we have $\sigma_i(a,d) = 1_{\{a_{i+1}+10-1+\sigma_{i+1}(a,d) \geq 10\}} = 1$ for all $i \in \{l(a)-1,...,1\}$ (Note that the $\sigma_i(a,d)$ are calculated in a decreasing order). Hence, the digits of x are then given by $x_1 = \mathcal{U}(a_1 + 0 + \sigma_1(a,d)) = \mathcal{U}(a_1 + 1) = a_1 + 1$ (since $a_1 \leq b-2$), $x_{l(x)} = x_{l(a)} = \mathcal{U}(a_{l(a)} + 10 - 1) = a_{l(a)} - 1$ (by lemma 2.1), and for all $i \in \{2,...,l(a)-1\}$, $x_i = \mathcal{U}(a_i + 10 - 1 + \sigma_i(a,d)) = \mathcal{U}(a_i + 10 - 1 + 1) = \mathcal{U}(a_i + 10) = a_i$ (also by lemma 2.1).

Hence, $x_1 + x_{l(x)} = a_1 + 1 + a_{l(a)} - 1 = a_1 + a_{l(a)}$ and since $x_i = a_i$ for all $i \in \{2,...,l(a)-1\}$, we then conclude that $l(\phi(x)) = l(\phi(a))$.

If $l(\phi(x)) = l(\phi(a)) = l(a) = l(x)$, then the digits of $\phi(x)$ are given as follows:

$\phi(x)_1 = \mathcal{U}(x_1 + x_{l(a)} + \epsilon_1(x)) = \mathcal{U}(x_1 + x_{l(a)} + 1_{\{x_2 + x_{l(a)-1} + \epsilon_2(x) + \geq 10\}}) = \mathcal{U}(a_1 + 1 + a_{l(a)} - 1 + 1_{\{x_2 + x_{l(a)-1} + \epsilon_2(x) \geq 10\}}) = \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2 + a_{l(a)-1} + \epsilon_2(a) \geq 10\}}) = \phi(a)_1$.

$\phi(x)_{l(x)} = \mathcal{U}(x_{l(x)} + x_1 + \epsilon_{l(x)}(x)) = \mathcal{U}(x_{l(a)} + x_1) = \mathcal{U}(a_{l(a)} - 1 + a_1 + 1) = \mathcal{U}(a_{l(a)} + a_1) = \phi(a)_{l(a)}$.

For all $i \in \{2, ..., l(a) - 1\}$,

$\phi(x)_i = \mathcal{U}(x_i + x_{l(a)-i+1} + \epsilon_i(x)) = \mathcal{U}(x_i + x_{l(a)-i+1} + 1_{\{x_{i+1} + x_{l(a)-i} + \epsilon_{i+1}(x) \geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i+1} + 1_{\{a_{i+1} + a_{l(a)-i} + \epsilon_{i+1}(x) \geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i} + 1_{\{a_{i+1} + a_{l(a)-i} + \epsilon_{i+1}(a) \geq 10\}}) = \phi(a)_i$.

If $l(\phi(x)) = l(\phi(a)) = l(a) + 1 = l(x) + 1$, then the digits of $\phi(x)$ are given as follows:

$\phi(x)_1 = 1 = \phi(a)_1$ because $x_1 + x_{l(x)} = a_1 + 1 + a_{l(a)} - 1 = a_1 + a_{l(a)}$ and $x_i = a_i$ for all $i \in \{2, ..., l(a) - 1\}$.

$\phi(x)_{l(x)+1} = \mathcal{U}(x_{l(x)} + x_1 + \epsilon_{l(x)}(x)) = \mathcal{U}(x_{l(a)} + x_1) = \mathcal{U}(a_{l(a)} - 1 + a_1 + 1) = \mathcal{U}(a_{l(a)} + a_1) = \phi(a)_{l(a)+1}$.

Since $x_1 + x_{l(x)} = a_1 + a_{l(a)}$ and $\epsilon_{l(x)}(x) = \epsilon_{l(a)}(a) = 0$

$\phi(x)_{l(x)} = \mathcal{U}(x_{l(x)-1} + x_2 + \epsilon_{l(x)-1}(x)) = \mathcal{U}(x_{l(x)-1} + x_2 + 1_{\{x_{l(x)} + x_1 + \epsilon_{l(x)}(x) \geq 10\}}) = \mathcal{U}(x_{l(x)-1} + x_2 + 1_{\{x_{l(x)} + x_1 \geq 10\}}) = \mathcal{U}(a_{l(a)-1} + a_2 + 1_{\{a_{l(a)} + a_1 \geq 10\}}) = \mathcal{U}(a_{l(a)-1} + a_2 + 1_{\{a_{l(a)} + a_1 + \epsilon_{l(a)}(a) \geq 10\}}) = \phi(a)_{l(a)}$.

$\phi(x)_2 = \mathcal{U}(x_1 + x_{l(a)} + \epsilon_1(x)) = \mathcal{U}(x_1 + x_{l(a)} + 1_{\{x_2 + x_{l(a)-1} + \epsilon_2(x) + \geq 10\}}) = \mathcal{U}(a_1 + 1 + a_{l(a)} - 1 + 1_{\{x_2 + x_{l(a)-1} + \epsilon_2(x) \geq 10\}}) = \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2 + a_{l(a)-1} + \epsilon_2(a) \geq 10\}}) = \phi(a)_2$.

For all $i \in \{3, ..., l(x) - 1\}$, we have $\{i - 1, l(x) - i + 2\} \subseteq \{2, ..., l(x) - 1\}$. Hence, $\phi(x)_i = \mathcal{U}(x_{i-1} + x_{l(x)-i+2} + \epsilon_{i-1}(x)) = \mathcal{U}(x_{i-1} + x_{l(x)-i+2} + 1_{\{x_i + x_{l(x)-i+1} + \epsilon_i(x) \geq 10\}}) = \mathcal{U}(a_{i-1} + a_{l(a)-i+2} + 1_{\{a_i + a_{l(a)-i+1} + \epsilon_i(x) \geq 10\}}) = \mathcal{U}(a_{i-1} + a_{l(a)-i+2} + 1_{\{a_i + a_{l(a)-i+1} + \epsilon_i(a) \geq 10\}}) = \phi(a)_i$. □

For example, let consider $a = 4009$ and $b = 10$. Then a verifies the hypothesis of proposition 3.1. $l(a) = 4$ and $10^{l(a)-1} - 1 = 10^3 - 1 = 1000 - 1 = 999$. Hence, $m = a + 10^{l(a)-1} - 1 = 4009 + 999 = 4009 + 0999 = 5008$. It then comes that $\phi(4009) = 4009 + 9004 = 13013$ and $\phi(5008) = 5008 + 8005 = 13013$, which means that $\phi(4009) = \phi(5008)$.

Corollary 3.1 *Let $a = a_1 a_2 ... a_{l(a)-1} a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If a is a Lychrel number such that $1 \leq a_1 \leq b - 2$ and $a_{l(a)} > 0$, then $a + 10^{l(a)-1} - 1$ is also a Lychrel number.*

Proof Let $a = a_1 a_2 ... a_{l(a)-1} a_{l(a)} \in N$ be a natural number. Since $a_1 \leq b - 2$ and $a_{l(a)} > 0$, then $\phi(a) = \phi(a + 10^{l(a)-1} - 1)$, from proposition 3.1. Hence, for all $n \in N$, $\phi^n(a) = \phi^n(a + 10^{l(a)-1} - 1)$. Since a is a Lychrel number, for all $n \in N$, $\phi^{n+1}(a) \neq 2\phi^n(a)$, which implies that $\phi^{n+1}(a + 10^{l(a)-1} - 1) \neq 2\phi^n(a + 10^{l(a)-1} - 1)$ for all $n \in N$. Hence, $a + 10^{l(a)-1} - 1$ is also a Lychrel number. □

Proposition 3.2 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If $l(a)$ is even, $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} > 0$, then $\phi(a) = \phi(a + (b-1).10^{\frac{l(a)}{2}-1})$.*

Proof Let consider such number $a = a_1a_2....a_{l(a)-1}a_{l(a)} \in N$. We set $x = a + (b-1).10^{\frac{l(a)}{2}-1}$. Then $l(x) = l(a)$ because $a_{\frac{l(a)}{2}} \leq b-2$ and $l((b-1).10^{\frac{l(a)}{2}-1}) = \frac{l(a)}{2}$. Moreover, setting $m = (b-1).10^{\frac{l(a)}{2}-1}$, $l(m) = \frac{l(a)}{2} \in N$ since $l(a)$ is even. We then consider d the number obtained by adding 0 to m $\frac{l(a)}{2}$ times and just before the first digit of m. Then $l(a) = l(d)$ and $a + m = a + d$. The digits of d are all equal to 0, except the one in the position $\frac{l(a)}{2}+1$ which is equal to $b-1$. Hence, we have $\sigma_{l(x)}(a,d) = 0$, $\sigma_i(a,d) = 1_{\{a_{i+1}+0+\sigma_{i+1}(a,d)\geq 10\}} = 0$ for all $i \in \{l(a)-1, ..., \frac{l(a)}{2}+1\}$ (Note that the $\sigma_i(a,d)$ are calculated in a decreasing order) and since $a_{\frac{l(a)}{2}+1} > 0$, $\sigma_{\frac{l(a)}{2}}(a,d) = 1$. Also, for all $i \in \{\frac{l(a)}{2}-1, ..., 1\}$, $\sigma_i(a,d) = 1_{\{a_{i+1}+0+\sigma_{i+1}(a,d)\geq 10\}} = 0$, since $a_{\frac{l(a)}{2}} \leq b-2$.

Hence, the digits of x are then expressed as the function of the digits of a as follows:
for all $i \in \{1, ..., \frac{l(a)}{2}-1\}$, $x_i = \mathcal{U}(a_i + 0 + \sigma_i(a,d)) = \mathcal{U}(a_i + 0) = a_i$.

$x_{\frac{l(a)}{2}} = \mathcal{U}(a_{\frac{l(a)}{2}} + 0 + \sigma_{\frac{l(a)}{2}}(a,d)) = \mathcal{U}(a_{\frac{l(a)}{2}} + 1) = a_{\frac{l(a)}{2}} + 1$ because $1 \leq a_{\frac{l(a)}{2}} \leq b-2$.

$x_{\frac{l(a)}{2}+1} = \mathcal{U}(a_{\frac{l(a)}{2}+1} + 10 - 1 + \sigma_{\frac{l(a)}{2}+1}(a,d)) = \mathcal{U}(a_{\frac{l(a)}{2}+1} + 10 - 1) + \sigma_{\frac{l(a)}{2}+1}(a,d)$(because of lemma 2.1)$= a_{\frac{l(a)}{2}+1} - 1 + \sigma_{\frac{l(a)}{2}+1}(a,d)$(also by lemma 2.1)$= a_{\frac{l(a)}{2}+1} - 1$.

for all $i \in \{\frac{l(a)}{2}+2, ..., l(a)\}$, $x_i = \mathcal{U}(a_i + 0 + \sigma_i(a,d)) = \mathcal{U}(a_i + 0) = a_i$.

From all the elements above, its comes that $i \in \{1, ..., l(x)\}$, $x_i + x_{l(x)-i+1} = a_i + a_{l(a)-i+1}$. We then conclude that $l(\phi(x)) = l(\phi(a))$. We can now calculate the digits of $\phi(x)$ as follows:

for all $i \in \{1, ..., \frac{l(a)}{2}-1\}$, $\frac{l(a)}{2}+2 \leq l(a)-i+1 \leq l(a)$. Hence, $\phi(x)_i = \mathcal{U}(x_i + x_{l(x)-i+1} + \epsilon_i(x)) = \mathcal{U}(x_i + x_{l(x)-i+1} + 1_{\{x_{i+1}+x_{l(x)-i}+\epsilon_{i+1}(x)\geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i+1} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}})$
$= \mathcal{U}(a_i + a_{l(a)-i+1} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}}) = \phi(a)_i$ (we precise that $x_{i+1} + x_{l(a)-i} = a_{i+1} + a_{l(a)-i}$ for all $i \in \{1, ..., \frac{l(a)}{2}-1\}$).

$\phi(x)_{\frac{l(x)}{2}} = \mathcal{U}(x_{\frac{l(x)}{2}} + x_{\frac{l(x)}{2}+1} + \epsilon_{\frac{l(x)}{2}}(x)) = \mathcal{U}(x_{\frac{l(a)}{2}} + x_{\frac{l(a)}{2}+1} + 1_{\{x_{\frac{l(a)}{2}+1}+x_{l(a)-\frac{l(a)}{2}}+\epsilon_{\frac{l(a)}{2}+1}(x)\geq 10\}})$
$= \mathcal{U}(a_{\frac{l(a)}{2}} + 1 + a_{\frac{l(a)}{2}+1} - 1 + 1_{\{a_{\frac{l(a)}{2}+1}+a_{l(a)-\frac{l(a)}{2}}+\epsilon_{\frac{l(a)}{2}+1}(a)\geq 10\}}(a)) = \phi(a)_{\frac{l(a)}{2}}$.

In the same way, $\phi(x)_{\frac{l(x)}{2}+1} = \mathcal{U}(x_{\frac{l(x)}{2}+1} + x_{\frac{l(x)}{2}} + \epsilon_{\frac{l(x)}{2}+1}(x)) = \mathcal{U}(a_{\frac{l(a)}{2}+1} - 1 + a_{\frac{l(a)}{2}} + 1 + 1_{\{a_{\frac{l(a)}{2}+2}+a_{\frac{l(a)}{2}-1}+\epsilon_{\frac{l(a)}{2}+2}(a)\geq 10\}}) = \phi(a)_{\frac{l(a)}{2}}$.

for all $i \in \{\frac{l(a)}{2}+2, ..., l(a)\}$, $x_i = \mathcal{U}(a_i + 0 + \sigma_i(a,d)) = \mathcal{U}(a_i + 0) = a_i$.

$\phi(x)_i = \mathcal{U}(x_i + x_{l(x)-i+1} + \epsilon_i(x)) = \mathcal{U}(x_i + x_{l(x)-i+1} + 1_{\{x_{i+1}+x_{l(x)-i}+\epsilon_{i+1}(x)\geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i+1} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}}) = \mathcal{U}(a_i +$

$a_{l(a)-i} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}})=\phi(a)_i$ (note that $x_{i+1} = a_{i+1}$ and $x_{l(a)-i} = a_{l(a)-i}$ for all $i \in \{\frac{l(a)}{2} + 2, ..., l(a)\}$). □

Considering the example where $a = 451400$ and $b = 10$. Then a verifies the hypothesis of proposition 3.2. Moreover, $a_1 = a + 9.10^{\frac{l(a)}{2}-1} = 451400 + 9.10^{\frac{6}{2}-1} = 451400 + 9.10^2 = 451400 + 900 = 451400 + 000900 = 452300$. $\phi(451400) = 451400 + 004154 = 455554$ and $\phi(452300) = 452300 + 003254 = 455554$. Hence, $\phi(451400) = \phi(452300)$.

Corollary 3.2 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If a is a Lychrel number, $l(a)$ is even, $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} > 0$, then $a + (b-1).10^{\frac{l(a)}{2}-1}$ is also a Lychrel number.*

Proof Let $a = a_1a_2....a_{l(a)-1}a_{l(a)} \in N$ be such natural number. Since $l(a)$ is even, $a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} > 0$, then $\phi(a) = \phi(a + (b-1).10^{\frac{l(a)}{2}-1})$, from proposition 3.2. Hence, for all $n \in N$, $\phi^n(a) = \phi^n(a + (b-1).10^{\frac{l(a)}{2}-1})$. Since a is a Lychrel number, for all $n \in N$, $\phi^{n+1}(a) \neq 2\phi^n(a)$, which implies that $\phi^{n+1}(a + (b-1).10^{\frac{l(a)}{2}-1}) \neq 2\phi^n(a + (b-1).10^{\frac{l(a)}{2}-1})$ for all $n \in N$. Hence, $a + (b-1).10^{\frac{l(a)}{2}-1}$ is also a Lychrel number. □

Proposition 3.3 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If $l(a)$ is even, $(a_1, a_{l(a)}) \neq (b-1, b-1)$, $a_1 + a_{l(a)} \geq 10$, $a_{\frac{l(a)}{2}-1} + a_{\frac{l(a)}{2}+2} \geq 10$, $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} = 0$, then $\phi^2(a) = \phi^2(a + (b-1).10^{\frac{l(a)}{2}-1})$.*

Proof Let consider such number $a = a_1a_2...a_{l(a)} \in N$. We set $x = a+(b-1).10^{\frac{l(a)}{2}-1}$. Then $l(x) = l(a)$ because $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $l((b-1).10^{\frac{l(a)}{2}-1}) = \frac{l(a)}{2}$. Moreover, setting $m = (b-1).10^{\frac{l(a)}{2}-1}$, $l(m) = \frac{l(a)}{2} \in N$ since $l(a)$ is even. We then consider d the number obtained by adding 0 to m $\frac{l(a)}{2}$ times and just before the first digit of m. Then $l(a) = l(d)$ and $a + m = a + d$. The digits of d are all equal to 0, except the one in position $\frac{l(a)}{2} + 1$ which is equal to $b-1$. Hence, we have $\sigma_{l(x)}(a, d) = 0$ and $\sigma_i(a, d) = 0$ for all $i \in \{l(a)-1, ..., 1\}\backslash\{\frac{l(a)}{2}\}$ (Note that the $\sigma_i(a, d)$ are calculated in a decreasing order) and since $a_{\frac{l(a)}{2}+1} = 0$, we also have $\sigma_{\frac{l(a)}{2}}(a, d) = 0$.

We then conclude that for all $i \in \{1, ..., l(x)\}\backslash\{\frac{l(a)}{2} + 1\}$, $x_i = \mathcal{U}(a_i + 0 + \sigma_i(a, d))=\mathcal{U}(a_i + 0) = a_i$. Moreover, $x_{\frac{l(a)}{2}+1} = \mathcal{U}(a_{\frac{l(a)}{2}+1} + 10 - 1 + \sigma_{\frac{l(a)}{2}+1}(a, d))=\mathcal{U}(0 + 10 - 1 + 0) = 10 - 1$.

Now we search for the digits of $\phi(x)$. In fact, $\phi(x) = \phi(x)_1\phi(x)_2.....\phi(x)_{l(\phi(x))}$. Since $x_1 + x_{l(x)} = a_1 + a_{l(a)} \geq 10$, then $l(\phi(x)) = l(x) + 1 = l(a) + 1$.

Hence, $\phi_1(x) = 1 = \phi_1(a)$ and $\phi(x)_{l(\phi(x))}=\mathcal{U}(x_{l(x)} + x_1)=\mathcal{U}(a_{l(a)} + a_1)=\phi(a)_{l(\phi(a))}$.

For all $i \in \{2, ..., \frac{l(a)}{2} - 1\}$, $\phi(x)_i=\mathcal{U}(x_{i-1} + x_{l(x)-i+2} + \epsilon_{i-1}(x))=\mathcal{U}(x_{i-1} + x_{l(x)-i+2} + 1_{\{x_i+x_{l(x)-i+1}+\epsilon_i(x)\geq 10\}})=\mathcal{U}(a_{i-1} + a_{l(a)-i+2} + 1_{\{a_i+a_{l(a)-i+1}+\epsilon_i(a)\geq 10\}})=\phi(a)_i$ (note that $x_{i+1} = a_{i+1}$ and $x_{l(a)-i} = a_{l(a)-i}$ for all $i \in \{2, ..., \frac{l(a)}{2} - 1\}$).

$\phi(x)_{\frac{l(x)}{2}}=\mathcal{U}(x_{\frac{l(x)}{2}-1} + x_{\frac{l(x)}{2}+2} + 1)$ (because $a_{\frac{l(a)}{2}} + 10 - 1 \geq 10$)$=\mathcal{U}(a_{\frac{l(a)}{2}-1} + a_{\frac{l(a)}{2}+2} + 1)=\mathcal{U}(a_{\frac{l(a)}{2}-1} + a_{\frac{l(a)}{2}+2}) + 1$ (because of lemma 2.1) $=\phi(a)_{\frac{l(a)}{2}} + 1$.

$\phi(x)_{\frac{l(x)}{2}+1}=\mathcal{U}(x_{\frac{l(x)}{2}} + 10 - 1 + 1)$(because $10 - 1 + a_{\frac{l(a)}{2}} \geq 10$)$=\mathcal{U}(a_{\frac{l(a)}{2}} + 10)=a_{\frac{l(a)}{2}}=\phi(a)_{\frac{l(a)}{2}+1}$.

$\phi(x)_{\frac{l(x)}{2}+2}=\mathcal{U}(10 - 1 + x_{\frac{l(x)}{2}} + \epsilon_{\frac{l(x)}{2}+1}(x))=\mathcal{U}(10 - 1 + a_{\frac{l(a)}{2}} + 1_{\{a_{\frac{l(a)}{2}+3}+a_{\frac{l(a)}{2}-2}+\epsilon_{\frac{l(a)}{2}+3}(a)\geq 10\}}(a))=a_{\frac{l(a)}{2}} - 1 + 1_{\{a_{\frac{l(a)}{2}+3}+a_{\frac{l(a)}{2}-2}+\epsilon_{\frac{l(a)}{2}+3}(a)\geq 10\}}$(because $1 \leq a_{\frac{l(a)}{2}} \leq b - 2$)$= \phi(a)_{\frac{l(a)}{2}+2} - 1$.

For all $i \in \{\frac{l(a)}{2} + 3, ..., l(a) + 1\}$, $\phi(x)_i=\mathcal{U}(x_{i-1} + x_{l(x)-i+2} + \epsilon_{i-1}(x))=\mathcal{U}(x_{i-1} + x_{l(x)-i+2} + 1_{\{x_i+x_{l(x)-i+1}+\epsilon_i(x)\geq 10\}})=\mathcal{U}(a_{i-1} + a_{l(a)-i+2} + 1_{\{a_i+a_{l(a)-i+1}+\epsilon_i(a)\geq 10\}})=\phi(a)_i$.

After finding the digits of $\phi(x)$, it is quite clear that $\phi(x) \neq \phi(a)$. Now we use the digits of $\phi(x)$ to determine that of $\phi^2(x)$.

In fact, $\phi(x)_1 + \phi(x)_{l(\phi(x))}=\phi(a)_1 + \phi(a)_{l(\phi(a))}=1 + \mathcal{U}(a_{l(a)} + a_1) \leq 8$ because $a_1 \neq 9 \neq a_{l(a)}$. Hence, $l(\phi^2(x)) = l(\phi(x)) = l(x) + 1 = l(a) + 1$.

It comes that for all $i \in \{1, ..., \frac{l(x)}{2} - 1\}$,
$\phi^2(x)_i=\mathcal{U}(\phi(x)_i + \phi(x)_{l(x)-i+1} + \epsilon_i(\phi(x)))=\mathcal{U}(\phi(x)_i + \phi(x)_{l(x)-i+1} + 1_{\{\phi(x)_{i+1}+\phi(x)_{l(x)-i}+\epsilon_{i+1}(\phi(x))\}})$
$=\mathcal{U}(\phi(a)_i + \phi(a)_{l(a)-i+1} + 1_{\{\phi(a)_{i+1}+a_{l(\phi(a))-i}+\epsilon_{i+1}(\phi(a))\}})=\phi^2(a)_i$.

Also, for all $i \in \{\frac{l(x)}{2} + 3, ..., l(x) + 1\}$,
$\phi^2(x)_i=\mathcal{U}(\phi(x)_i + \phi(x)_{l(x)-i+1} + \epsilon_i(\phi(x)))=\mathcal{U}(\phi(x)_i + \phi(x)_{l(x)-i+1} + 1_{\{\phi(x)_{i+1}+\phi(x)_{l(x)-i}+\epsilon_{i+1}(\phi(x))\}})$
$=\mathcal{U}(\phi(a)_i + \phi(a)_{l(a)-i+1} + 1_{\{\phi(a)_{i+1}+\phi(a)_{l(\phi(a))-i}+\epsilon_{i+1}(\phi(a))\}})=\phi^2(a)_i$.

For $i = \frac{l(x)}{2}$,
$\phi^2(x)_{\frac{l(x)}{2}}=\mathcal{U}(\phi(x)_{\frac{l(x)}{2}} + \phi(x)_{\frac{l(x)}{2}+2} + \epsilon_{\frac{l(x)}{2}}(\phi(x)))=\mathcal{U}(\phi(a)_{\frac{l(a)}{2}} + 1+\phi(a)_{\frac{l(a)}{2}} - 1+ 1_{\{\phi(a)_{\frac{l(a)}{2}+1}+\phi(a)_{\frac{l(a)}{2}-1}+\epsilon_{\frac{l(a)}{2}+1}(\phi(a))\}})=\mathcal{U}(\phi(a)_{\frac{l(a)}{2}} + \phi(a)_{\frac{l(a)}{2}} + 1_{\{\phi(a)_{\frac{l(a)}{2}+1}+\phi(a)_{\frac{l(a)}{2}-1}\epsilon_{\frac{l(a)}{2}+1}(\phi(a))\}}) =$
$\phi^2(a)_{\frac{l(a)}{2}}$.

In the same way, $\phi^2(x)_{\frac{l(a)}{2}+2}=\phi^2(a)_{\frac{l(a)}{2}+1}$.

For $i = \frac{l(x)}{2} + 1$,
$\phi^2(x)_{\frac{l(x)}{2}+1}=\mathcal{U}(\phi(x)_{\frac{l(x)}{2}+1} + \phi(x)_{\frac{l(x)}{2}+1} + \epsilon_{\frac{l(x)}{2}+1}(\phi(x)))=\mathcal{U}(\phi(a)_{\frac{l(a)}{2}} + 1+\phi(a)_{\frac{l(a)}{2}} - 1+ 1_{\{\phi(x)_{\frac{l(x)}{2}+1}+\phi(x)_{\frac{l(x)}{2}-1}+\epsilon_{\frac{l(x)}{2}+1}(\phi(x))\}})=\mathcal{U}(2.\phi(a)_{\frac{l(a)}{2}} + 1_{\{\phi(a)_{\frac{l(a)}{2}+2}+\phi(a)_{\frac{l(a)}{2}}+\epsilon_{\frac{l(a)}{2}+2}(\phi(a))\}})=\phi^2(a)_{\frac{l(a)}{2}+1}$. □

We consider the natural number $a = 752069$ and $b = 10$. Then a verifies the hypothesis of proposition 3.3. Moreover, $a_1 = a + 9.10^{\frac{l(a)}{2}-1} = 752069 + 9.10^{\frac{6}{2}-1} = 752069 + 9.10^2 = 752069 + 900 = 752069 + 000900 = 752969$. $\phi(752069) = 752069 + 960257 = 1712326$ and $\phi(752969) = 752969 + 969257 = 1722226$. Hence, $\phi(752069) \neq \phi(752969)$. But $\phi^2(752069) = \phi(1712326) = 1712326 + 6232171 = 7944497$ and $\phi^2(752969) = \phi(1722226) = 1722226 + 6222271 = 7944497$. We then conclude that $\phi^2(752069) = \phi^2(752969)$.

Corollary 3.3 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 2$. If a is a Lychrel number, $l(a)$ is even, $(a_1, a_{l(a)}) \neq (b-1, b-1)$, $a_1 + a_{l(a)} \geq 10$, $a_{\frac{l(a)}{2}-1} + a_{\frac{l(a)}{2}+2} \geq 10$, $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} = 0$, then $a + (b-1).10^{\frac{l(a)}{2}-1}$ is also a Lychrel number.*

Proof Let $a = a_1a_2...a_{l(a)} \in N$ be a natural number. Since $l(a)$ is even, $a_1 + a_{l(a)} \geq 10$, $a_{\frac{l(a)}{2}-1} + a_{\frac{l(a)}{2}+2} \geq 10$, $1 \leq a_{\frac{l(a)}{2}} \leq b-2$ and $a_{\frac{l(a)}{2}+1} = 0$, then $\phi^2(a) = \phi^2(a + (b-1).10^{\frac{l(a)}{2}-1})$, from proposition 3.3. Hence, for all $n \in N\backslash\{0,1\}$, $\phi^n(a) = \phi^n(a+(b-1).10^{\frac{l(a)}{2}-1})$. Since a is a Lychrel number, for all $n \in N$, $\phi^{n+1}(a) \neq 2\phi^n(a)$, which implies that $\phi^{n+1}(a + (b-1).10^{\frac{l(a)}{2}-1}) \neq 2\phi^n(a + (b-1).10^{\frac{l(a)}{2}-1})$ for all $n \in N$. Hence, $a + (b-1).10^{\frac{l(a)}{2}-1}$ is also a Lychrel number. □

Now what happen when the natural number $l(a)$ is odd?.

Let $a = a_1a_2....a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b $(3 \leq b \leq 10)$ such that $l(a)$ is odd. We consider the natural number $m = \sum_{k=1}^{l(a)-3}(b-1).10^k$. For example, if $b = 10$ and $l(a) = 5$, $m = \sum_{k=1}^{5-3} 9.10^k = \sum_{k=1}^{2} 9.10^k = 9.10^1 + 9.10^2 = 990$. Then by construction, $l(m) = l(a) - 2$ and all de digits of c are all equal to $10-1$, except the last digit which is equal to 0. Moreover, considering d the number obtained by adding 0 to m twice and just before the first digit of m, we have $l(a) = l(d)$ and $a + m = a + d$. The digits of d are all equal to $b-1$, except the first, the second and the last which are equal to 0.

We suppose that $a_{l(a)-1} \neq 0$. Then setting $x = a+d$, $l(x) = l(a)$ if and only if $a_1 \neq b-1$ and $a_2 \neq b-1$. Hence, we have $\sigma_{l(a)}(a,d) = 0$ by definition of σ, and $\sigma_{l(a)-1}(a,d) = 0$ because $a_{l(a)} + 0 \leq b-1$. Also, $\sigma_1(a,d) = 0$ because $a_2 \neq b-1$, and for all $i \in \{2, ..., l(a)-2\}$, $\sigma_i(a,d) = 1$ since $a_{i+1} + 10 - 1 + \sigma_{i+1}(a,d) \geq 10$ (we remind that the the $\sigma_i(a,d)$ are determined in the decreasing order). From the $\sigma_i(a,d)$, the digits of x are now given by $x_1 = \mathcal{U}(a_1 + \sigma_1(a,d)) = a_1$ since $a_1 \neq b-1$ and $\sigma_1(a,d) = 0$. $x_2 = \mathcal{U}(a_2 + \sigma_2(a,d)) = \mathcal{U}(a_2 + 1) = a_2 + 1$ since $a_2 \neq b-1$. Moreover, for all $i \in \{3, ..., l(a)-2\}$, $x_i = \mathcal{U}(a_i + b - 1 + \sigma_i(a,d)) = \mathcal{U}(a_i + b - 1 + 1) = \mathcal{U}(a_i + b) = \mathcal{U}(a_i + 10) = a_i$ (by lemma 2.1). Also, $x_{l(a)-1} = \mathcal{U}(a_{l(a)-1} + b - 1 + \sigma_{l(a)-1}(a,d)) = \mathcal{U}(a_{l(a)-1} + b - 1) = \mathcal{U}(a_{l(a)-1} + 10 - 1) = a_{l(a)-1} - 1$ (also by lemma 2.1) and $x_{l(a)} = \mathcal{U}(a_{l(a)} + 0 + \sigma_{l(a)}(a,d)) = \mathcal{U}(a_{l(a)} + 0 + 0) = a_{l(a)}$. We now determine the digits of $\phi(x)$ using the digits of x as follow.

for all $i \in \{3, ..., l(a) - 2\}$, $\{l(a) - i + 1, l(a) - 1\} \subseteq \{2, ..., l(a) - 2\}$, and then $x_i = a_i$ and $x_{l(x)-i+1} = a_{l(a)-i+1}$. Hence, $x_i + x_{l(x)-i+1} = a_i + a_{l(a)-i+1}$. Moreover $x_2 + x_{l(x)-1} = a_2 + 1 + a_{l(a)-1} - 1 = a_2 + a_{l(a)-1}$ and $x_1 + x_{l(x)} = a_1 + a_{l(a)}$. We then conclude that $l(\phi(x)) = l(\phi(a))$. The digits of $\phi(x)$ can then be determined as follow:

If $l(\phi(x)) = l(\phi(a)) = l(a)$,

$\phi(x)_{l(\phi(x))} = \mathcal{U}(x_{l(x)} + x_1 + \epsilon_{l(x)}(x)) = \mathcal{U}(x_{l(a)} + x_1) = \mathcal{U}(a_{l(a)} + a_1) = \mathcal{U}(a_{l(a)} + a_1) = \phi(a)_{l(a)}$.

$\phi(x)_{l(\phi(x))-1} = \mathcal{U}(x_{l(x)-1} + x_2 + \epsilon_{l(x)-1}(x)) = \mathcal{U}(x_{l(a)-1} + x_2 + 1_{\{x_{l(a)}+x_2+\epsilon_{l(x)}(x)\}}) = \mathcal{U}(a_{l(a)-1} - 1 + a_2 + 1 + 1_{\{x_{l(a)}+x_2\}}) = \mathcal{U}(a_{l(a)-1} + a_2) = \phi(a)_{l(a)-1}$.

$\phi(x)_1 = \mathcal{U}(x_1 + x_{l(x)} + \epsilon_1(x)) = \mathcal{U}(x_1 + x_{l(x)} + 1_{\{x_2+x_{l(x)-1}+\epsilon_2(x)\geq 10\}}) = \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2+a_{l(a)-1}+\epsilon_2\geq 10\}})$
$= \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2+a_{l(a)-1}+\epsilon_2\geq 10\}}(a)) = \phi(a)_1$.

$\phi(x)_2 = \mathcal{U}(x_2 + x_{l(x)-1} + \epsilon_2(x)) = \mathcal{U}(x_2 + x_{l(x)-1} + 1_{\{x_3+x_{l(x)-2}+\epsilon_3(x)\geq 10\}}) = \mathcal{U}(a_2 + 1 + a_{l(a)-1} - 1 + 1_{\{a_3+a_{l(a)-2}+\epsilon_3(a)\geq 10\}}) = \mathcal{U}(a_2 + a_{l(a)-1} + 1_{\{a_3+a_{l(a)-2}+\epsilon_3(a)\geq 10\}}) = \phi(a)_2$.

For all $i \in \{3, ..., l(a) - 2\}$,
$\phi(x)_i = \mathcal{U}(x_i + x_{l(x)-i+1} + \epsilon_i(x)) = \mathcal{U}(x_i + x_{l(x)-i+1} + 1_{\{x_{i+1}+x_{l(x)-i}+\epsilon_{i+1}(x)\geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i+1} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}}) = \mathcal{U}(a_i + a_{l(a)-i} + 1_{\{a_{i+1}+a_{l(a)-i}+\epsilon_{i+1}(a)\geq 10\}}) = \phi(a)_i$.

If $l(\phi(x)) = l(\phi(a)) = l(a) + 1$,

$\phi(x)_1 = 1 = \phi(a)_1$ by definition.

$\phi(x)_2 = \mathcal{U}(x_1 + x_{l(x)} + \epsilon_1(x)) = \mathcal{U}(x_1 + x_{l(x)} + 1_{\{x_2+x_{l(x)-1}+\epsilon_2(x)\geq 10\}}) = \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2+a_{l(a)-1}+\epsilon_2(a)\geq 10\}}) = \mathcal{U}(a_1 + a_{l(a)} + 1_{\{a_2+a_{l(a)-1}+\epsilon_2(a)\geq 10\}}) = \phi(a)_2$.

$\phi(x)_3 = \mathcal{U}(x_2 + x_{l(x)-1} + \epsilon_2(x)) = \mathcal{U}(x_2 + x_{l(x)-1} + 1_{\{x_3+x_{l(x)-2}+\epsilon_3(x)\geq 10\}}) = \mathcal{U}(a_2 + 1 + a_{l(a)-1} - 1 + 1_{\{a_3+a_{l(a)-2}+\epsilon_3(a)\geq 10\}}) = \mathcal{U}(a_2 + a_{l(a)-1} + 1_{\{a_3+a_{l(a)-2}+\epsilon_3(a)\geq 10\}}) = \phi(a)_3$.

For all $i \in \{4, ..., l(a) - 2\}$, $\{i - 1, l(a) - i + 2\} \subseteq \{3, ..., l(a) - 2\}$ and then $x_{i-1} = a_{i-1}$ and $x_{l(a)-i+2} = a_{l(a)-i+2}$
$\phi(x)_i = \mathcal{U}(x_{i-1} + x_{l(x)-i+2} + \epsilon_i(x)) = \mathcal{U}(x_{i-1} + x_{l(x)-i+2} + 1_{\{x_i+x_{l(x)-i+1}+\epsilon_i(x)\geq 10\}}) = \mathcal{U}(a_{i-1} + a_{l(a)-i+2} + 1_{\{a_i+a_{l(a)-i+1}+\epsilon_i(a)\geq 10\}}) = \mathcal{U}(a_{i-1} + a_{l(a)-i+1} + 1_{\{a_i+a_{l(a)-i+1}+\epsilon_i(a)\geq 10\}}) = \phi(a)_i$.

$\phi(x)_{l(\phi(x))} = \mathcal{U}(x_{l(x)} + x_1 + \epsilon_{l(x)}(x)) = \mathcal{U}(x_{l(a)} + x_1) = \mathcal{U}(a_{l(a)} + a_1) = \mathcal{U}(a_{l(a)} + a_1) = \phi(a)_{l(a)}$.

$\phi(x)_{l(\phi(x))-1} = \mathcal{U}(x_{l(x)-1} + x_2 + \epsilon_{l(x)-1}(x)) = \mathcal{U}(x_{l(a)-1} + x_2 + 1_{\{x_{l(a)}+x_2+\epsilon_{l(x)}(x)\}}) = \mathcal{U}(a_{l(a)-1} - 1 + a_2 + 1 + 1_{\{x_{l(a)}+x_2\}}) = \mathcal{U}(a_{l(a)-1} + a_2) = \phi(a)_{l(a)-1}$.

That leads us to the following result:

Proposition 3.4 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$ and $l(a) \geq 5$. If $l(a)$ is odd, $a_1 \neq b - 1$, $a_2 \neq b - 1$ and $a_{l(a)-1} \neq 0$, then $\phi(a) = \phi(a + \sum_{k=1}^{l(a)-3}(b - 1).10^k)$.*

For example, we consider $a = 88010$ and $b = 10$. Then a verifies the hypothesis of proposition 3.4. Moreover, $a_1 = a + \sum_{k=1}^{l(a)-3} 9.10^k = 88010 + \sum_{k=1}^{5-3} 9.10^k = 88010 + 9.10^2 + 9.10^1 = 88010 + 900 + 90 = 88010 + 990 = 88010 + 00990 = 89000$. $\phi(88010) = 88010 + 01088 = 89098$ and $\phi(89000) = 89000 + 00098 = 89098$. Hence, $\phi(88010) = \phi(89000)$.

Corollary 3.4 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)} \in N$ be a natural number in base b such that $3 \leq b \leq 10$, $l(a) \geq 5$, $l(a)$ is odd, $a_1 \neq b-1$, $a_2 \neq b-1$ and $a_{l(a)-1} \neq 0$. If a is a Lychrel number, then $a + \sum_{k=1}^{l(a)-3}(b-1).10^k$ is also a Lychrel number.*

Proof Let $a \in N$ be such natural number. Then $\phi(a) = \phi(a + \sum_{k=1}^{l(a)-3}.(b-1).10^k)$, from proposition 3.4. Hence, for all $n \in N$, $\phi^n(a) = \phi^n(a + \sum_{k=1}^{l(a)-3}.(b-1).10^k)$. Since a is a Lychrel number, for all $n \in N$, $\phi^{n+1}(a) \neq 2\phi^n(a)$, which implies that $\phi^{n+1}(a+\sum_{k=1}^{l(a)-3}.(b-1).10^k) \neq 2\phi^n(a+\sum_{k=1}^{l(a)-3}.(b-1).10^k)$ for all $n \in N$. Hence, $a + \sum_{k=1}^{l(a)-3}.(b-1).10^k$ is also a Lychrel number. □

Remark 3.1 *The above corollary 3.1, corollary 3.2, corollary 3.3 and corollary 3.4 also hold for Lychrel candidates.*

4 Applying the above properties on some cases

In this section, we studies two cases. The first case is the one where $b = 4$, and we try to look for a Lychrel number in this base and to apply the above properties on this Lychrel number to obtained others. The second one is the case where $b = 10$.

Case 1: Let suppose that $b = 4$ and let consider $a = 255$ (in base 10). When we convert a in base 4, we obtain $a = 3333$.

Then in base 4, $\phi^6(a) = 103323000$. Since $33230 = 10.(3.10^3 + 323)$, we replace 33230 in the expression of $\phi^6(a)$ by the expression $[10.(3.10^3 + 323)]$ and finally, $\phi^6(a)$ is written as follow: $\phi^6(a) = 10[10.(3.10^3 + 323)]00$.

We set
$x_0 = 323$,
$y_0 = 10[x_0]00$, (which is equal to $10[323]00 = 1032300$)
and for all $p \geq 1$,
$x_{2p-1} = 10^p.x_0 + 3.10^{l(x_0)+p}(\sum_{i=0}^{p-1} 10^i)$,
$x_{2p} = 10.x_{2p-1}$,
$y_{2p-1} = 10[x_{2p-1}]00$ and
$y_{2p} = 10[x_{2p}]00$.

Then for $p = 1$, $\phi^3(y_1) = \phi^6(a)$ and for all $p \geq 1$, $\phi^3(y_{2p-1}) = y_{2p}$ and $\phi^3(y_{2p}) = y_{2p+1)}$, which means that the iterations of a contain infinite non

palindromic sequences (since no y_i is a palindrome). Moreover, from the construction of the x_p and y_p, it can be noticed that $\phi(y_{2p-1})$,$\phi^2(y_{2p-1})$,$\phi(y_{2p})$ and $\phi^2(y_{2p})$ are not palindromes. These make $a = 255$ a Lychrel number in base $b = 4$.

Note that a is not a Lychrel Number in base 10 because $\phi^3(255) = 6666$ which is a palindrome.

Other non trivial Lychrel numbers can be obtained from a using the properties defined in the previous section.

Hence, $\phi(3333) = 13332 = a_1$. a_1 is trivially a Lychrel number in base 4, and verifies the hypothesis of proposition 3.1. Therefore, since a_1 is a Lychrel Number, $a_2 = a_1 + 10^{l(a_1)-1} - 1 = 13332 + 10^{5-1} - 1 = 23331$ is also a Lychrel number because they belong to the same equivalence class of $\mathcal{R}_5$.

Moreover, $a_3 = \phi^3(3333) = 1033230$. It is a Lychrel number and verifies the hypothesis of proposition 3.4. Therefore, $a_4 = a_3 + \sum_{k=1}^{l(a_3)-3}(b-1).10^k = 1033230 + \sum_{k=1}^{7-3}(4-1).10^k = 1033230 + 33330 = 1133220$ is also a Lychrel number because they belong to the same equivalence class of $\mathcal{R}_7$.

One can observe with the above case that a palindrome 3333 is also a Lychrel number in base 4. We call this type of numbers **Palindrome-Lychrel numbers**. In the list of Lychrel candidates base 10 from (10), Palindrome-Lychrel candidates are not considered, even though their existence was partially mentioned in (11) where Lychrel candidates below 10000 are computed, but with a threshold of only 50 iterations. They were also implicitly mentioned in (6), with a threshold of 100 iterations. Note that on Palindrome-Lychrel numbers, when we apply the iterative process of reversing their digits and adding their reverse numbers, we will never return to a palindrome. However, a palindrome which is not a Palindrome-Lychrel number will lead to another palindrome after the application of the iterative process of reversing its digits and adding its reverse number.

Case 2: Let suppose that $b = 10$. Here we have not been able to exhibit a Lychrel number despite all our effort. However, we apply the found properties on candidates.

In base 10, when we take the first Lychrel candidate **196** and iteratively apply proposition 3.1 on it, we obtain the other elements of the first column of the table bellow, from 295 to 790 and we conclude from corollary 3.1 that they too are Lychrel candidates. Also, considering the first iteration of 196 which is 887 and iteratively applying the same result, we obtain the numbers 689, 788 and 986 and conclude from the same corollary that they too are Lychrel candidates. The same situation occurs for 879 and we obtain 978. Hence, with 196, $\phi(196) = 887$ and 879, we obtain all the other elements of the table below.

196	689
295	788
394	**887**
493	986
592	879
691	978
790	

Now we take the second iteration of candidate 196, which is **1675**, and we iteratively apply proposition 3.1 on it to obtained the elements of the third column of the table below, which are 2674, 3673, 4672, 5671 and 6670. We then conclude from from corollary 3.1 that they are also Lychrel candidates. On each elements of this third column, we iteratively apply proposition 3.2 to obtain from each of them all the other elements of the corresponding line. We then conclude from corollary 3.1 that they too are Lychrel candidates. Hence, all the 35 other elements of this tables are obtained just from a single element which is 1675.

1495	1585	**1675**	1765	1855	1945
2494	2584	2674	2764	2854	2944
3493	3583	3673	3763	3853	3943
4492	4582	4672	4762	4852	4942
5491	5581	5671	5761	5851	5941
6490	6580	6670	6760	6850	6940

The same process is applied when we consider the Lychrel candidate **1857** which is the first iteration of 879 and **1997** as we can see in the following tables:

1497	1587	1677	1767	**1857**	1947
2496	2586	2676	2766	2856	2946
3495	3585	3675	3765	3855	3945
4494	4584	4674	4764	4854	4944
5493	5583	5673	5763	5853	5943
6492	6582	6672	6762	6852	6942
7491	7581	7671	7761	7851	7941
8490	8580	8670	8760	8850	8940

1997
2996
3995
4994
5993
6992
7991
8990

Remark 4.1 *The elements of the last table above are iteratively obtained from* 1997 *by applying proposition 3.1. In this table, it can be seen that one of the obtained element,* ***4994*** *is a Palindrome. this number is then a Palindrome-Lychrel candidate.*

4079	4169	4259	4349	4439	4529	4619	4709	4799	4889	4979
5078	5168	5258	5348	5438	5528	5618	5708	5798	5888	5978
6077	6167	6257	6347	6437	6527	6617	6707	6797	6887	6977
7076	7166	7256	7346	**7436**	7526	7616	7706	7796	7886	7976
8075	8165	8255	8345	8435	8525	8615	8705	8795	8885	8975
9074	9164	9254	9344	9434	9524	9614	9704	9794	9884	9974

Considering the third iteration of 196 which is 7436 which, when we iteratively apply proposition 3.1 to it we obtain the elements of the fifth column of the above table. The elements of this column are Lychrel candidate, from corollary 3.1. Moreover when we iteratively apply proposition 3.2 to each element of this column we obtain the elements from column 1 to column 8 (except column 5). The 48 elements of the white part of the table belong to the same equivalence class of the relation $\mathcal{R}_4$ having 7436 as the representative, and are

Lychrel candidates from corollary 3.2. Now we consider column 8 of the (white part of the) table and iteratively apply proposition 3.3 to each element of this column because they verify the hypothesis of this proposition. We then obtain the first column of the green part of the table and again iteratively apply proposition 3.2 to each element of this column to obtain the elements of the second and the third column of the green part of the table. Note that all the 18 elements of the green part of the table also belong to the same equivalence class which is different from the one containing the elements of the white part of the table and are Lychrel candidate from corollary 3.3. Hence, all these 66 element of this tables are Lychrel candidates.

The same situation occur with the table below where starting with the Lychrel number 9438 which is the second iteration of 879, and following the same steps as the previous cases. The white part of this table is also an equivalence class of the relation $\mathcal{R}_4$. proposition 3.3 is also iteratively applied to each element of column 8 to obtain the elements of the first columns of the blue part, and proposition 3.2 is iteratively applied to each element of this column to obtain the other columns of the blue part of the table. Also, all the 22 elements are Lychrel candidates.

8079	8169	8259	8349	8439	8529	8619	8709	8799	8889	8979
9078	9168	9258	9348	**9438**	9528	9618	9708	9798	9888	9978

The following tables are obtained in the same way. In the first one, we start with the Lychrel candidate 9988 which is the first iteration of 1997 and iteratively apply proposition 3.1 to it to obtain the elements of its column. After that, proposition 3.2 is applied to obtained the other column of the blue part of the table. After, proposition 3.3 is then applied to the tenth column to obtain the ninth column of the table. To end, proposition 3.2 is iteratively applied to the elements of column 9 to obtain the other elements of the white part of the table.

8089	8179	8269	8359	8449	8539	8629	8719	8809	8899	8989
9088	9178	9268	9358	9448	9538	9628	9718	9808	9898	**9988**

However, the second table (below) seems not to be link to any iteration of a Lychrel candidate. The smallness element of this table is 7059 and it contains another palindrome-Lychrel candidate which is 8778. Hence, proposition 3.1 is applied to 7059 to obtain the elements of its column, which are also candidates from corollary 3.1. After that, proposition 3.2 is applied to each element of this column to obtain the other column of the pink part of the table, which are candidates too, by corollary 3.2. After, proposition 3.3 is applied to each element of column 7 to obtain the elements of column 6 and at the end, proposition 3.2 is iteratively applied to each element of column 6 to obtain the other column of the white part of the table. Note that the White part of the table is an equivalent class and the pink part of the table is another equivalent class of $\mathcal{R}_4$. All the elements of this tables are Lychrel candidates.

7059	7149	7239	7329	7419	7509	7599	7689	7779	7869	7959
8058	8148	8238	8328	8418	8508	8598	8688	**8778**	8868	8958
9057	9147	9237	9327	9417	9507	9597	9687	9777	9867	9957

Remark 4.2 *One can observe that with the only four following Lychrel candidates base* 10*,* 196*,* 879*,*1997 *and* 7059*, one can obtain all the* 248 *known Lychrel candidates base* 10 *less than* 10000*, by iteratively applying the properties defined in the previous section.*

The defined properties also enable us to describe Lychrel candidates above 10000 as it is shown in the table below. Hence from 13783 which is the fifth iteration of 196, and applying proposition 3.1, we obtain the numbers in the second column of the table which are Lychrel candidates by corollary 3.1. After that, we apply proposition 3.4 on each element of this column to obtained all the elements of the other columns which are also Lychrel candidates. Notes that this table is also an equivalence class of the relation $\mathcal{R}_5$.

12793	**13783**	14773	15763	16753	17743	18733	19723
22792	23782	24772	25762	26752	27742	28732	29722
32791	33781	34771	35761	36751	37741	38731	39721
42790	43780	44770	45760	46750	47740	48730	49720

Also, considering the Lychrel candidates **17787** and **18887** which are respectively the third iteration of 879 and the second iteration of 1997, and applying proposition 3.1 and proposition 3.4 on each of them with the same process as the previous case, we obtain the following tables of new candidates.

16797	**17787**	18777	19767
26796	27786	28776	29766
36795	37785	38775	39765
46794	47784	48774	49764
56793	57783	58773	59763
66792	67782	68772	69762
76791	77781	78771	79761
86790	87780	88770	89760

17897	**18887**	19877
27896	28886	29876
37895	38885	39875
47894	48884	49874
57893	58883	59873
67892	68882	69872
77891	78881	79871
87890	88880	89870

One can observe that the number **18887** yields another Kuitché candidate in the last table above, which is **48884**. Note that the two tables above are equivalent classes of $\mathcal{R}_5$. It is also the case with the following table from the first iteration of the Lychrel number 7059, which is **16566**.

13596	14586	15576	**16566**	17556	18546	19536
23595	24585	25575	26565	27555	28545	29535
33594	34584	35574	36564	37554	38544	39534
43593	44583	45573	46563	47553	48543	49533
53592	54582	55572	56562	57552	58542	59532
63591	64581	65571	66561	67551	68541	69531
73590	74580	75570	76560	77550	78540	79530

In the next section we do the test of some of the new numbers found in base 10 to effectively confirm that they are Lychrel or Palindrome-Lychrel candidates.

Remark 4.3 *Under the hypothesis of corollary 3.1, corollary 3.2, corollary 3.3 and corollary 3.4, if the natural number a is not a Lychrel number, then respectively, $a+10^{l(a)-1}-1$, $a+(b-1).10^{\frac{l(a)}{2}-1}$ and $a+\sum_{k=1}^{l(a)-3}.(b-1).10^{k}$ will not be Lychrel numbers or Palindrome-Lychrel numbers.*

5 Test of some new found Lychrel candidates and Palindrome-Lychrel candidates in base 10

As we have expose in the introduction, several candidates has been tested by many authors in order to confirm whether they are Lychrel candidate, notably for natural numbers below 10000. Our aim is to test some of the numbers of the last four tables of the previous section, with a relatively high threshold, let say more than 100000 iterations. The choice of the numbers of these tables comes from the fact that they are above 10000 and then they have not been probably tested. However, for each table, there is no need testing all the elements of these tables because each of them is an equivalent class of the relation $\mathcal{R}_5$, and their elements will therefore generate the same output for the same number of iterations, especially for high thresholds. Hence, with specified thresholds, we have randomly chosen to test the following numbers 19723, 89760, 58883 and 43593, one from each class, and the following palindromes 4994 and 8778 from these tables.

For the test to be well done, we have written a code in Anaconda Navigator software, which takes a natural number, executes the iterative process of reversing and adding the reverse number, returns its palindrome if the threshold is not reached, and the last iteration (the one corresponding to the threshold) and its length if the threshold is reached without a palindrome been obtained. In this last case, for a very high threshold (more than 100000 iterations), the input number is concluded to be a Lychrel candidate or a Palindrome-Lychrel candidate. The code can be seen in the figure below:

```
i=0
o=0
num=4994
rev_num_str=str(num)[::-1]
if type(num)==float:
    reverse_num=float(rev_num_str)
elif type(num)==int:
    reverse_num=int(rev_num_str)
nu=num+reverse_num
while (o!=nu) and (i!=250000):
    i=i+1
    nu=nu+o
    rev_num_str=str(nu)[::-1]
    if type(nu)==float:
        reverse_num=float(rev_num_str)
    elif type(nu)==int:
        reverse_num=int(rev_num_str)
    o=reverse_num
print(nu)
print(len(str(nu)))
print(i)
```

The code is globally conceived in a R.A.I (Reverse, Add and Iterate) process, and is therefore described in tree mean steps: firstly, a natural input number (*num*) is taken and reversed. Secondly, the reverse number obtained (*reverse-num*) is added to the input number and the first iteration (*nu*) is obtained. These two steps are described from line 1 to line 9 of the above code. Thirdly, the process described in the first and the second steps is iterated β times (threshold of the iteration), from line 10 to line 18 of the code. At the end, if a palindrome is obtained, then the program stops and returns this palindrome (*nu*), its length (*len(str(nu))*) and the number of iterations been computed (i). If the threshold is reached without a palindrome been obtained, then the βth iteration is returned as output, its length and the threshold (the last iteration computed). The tested number is therefore considered as a Lychrel candidate or Palindrome-Lychrel candidate.

The test of the six numbers to be tested 19723, 89760, 58883, 43593, 4994 and 8778 is progressive, with different thresholds.

For $\beta = 100000$ and $\beta = 250000$, the result appears in the following table.

Number	8 778	4 994	19 723	89 760	58 883	43 593
β (Threshold)	100 000	250 000	250 000	250 000	250 000	250 000
Time took by the program to reach the Threshold iteration	1h	24h	24h	24h	24h	24h
First ten digits of the βth iteration	1772866159	1765322217	9052771407	9954280876	1574808897	9422139596
Ten last digits of the βth iteration	8951558287	1722223566	6932871508	6889824688	8688917385	6960412238
Number of digits of the βth iteration	41 436	103 464	103 566	103 622	103 465	103 706

One can see that with the use of each threshold $\beta \in \{100000, 250000\}$, a palindrome is not reached after the process.

We then increase the threshold for more than 500000 and carry out the test again. The table below presents the results obtained:

Number	8 778	4 994	19 723	89 760	58 883	43 593
β (Threshold)	530 345	531 067	530 914	530 520	531 051	530 404
Time took by the program to reach the Threshold iteration	8 days	8 days	8 days	8 days	8 days	8 days
First ten digits of the βth iteration	1026066438	9128795297	9432996208	8539096138	6617736506	1054105771
Ten last digits of the βth iteration	7844660520	7034988208	7917003338	9406009447	6046377166	3177502549
Number of digits of the βth iteration	220 013	219 799	219 927	219 969	219 793	220 037

For each of the proposed numbers, with a threshold more than 500000, a palindrome is still not reached.

We then conclude that 19723, 89760, 58883 and 43593 are effectively Lychrel candidates, and 4994 and 8778 are effectively Palindrome-Lychrel candidates.

6 What about the Palindrome-Lychrel numbers?

The present section addresses more deeply the notion of Palindrome-Lychrel numbers and explains how to obtain them.

Let consider a Lychrel number $a = a_1a_2...a_{l(a)-1}a_{l(a)}$ in base b such that $3 \leq b \leq 10$. If a Palindrome-Lychrel number $k(a)$ can be obtained from a, then $k(a)$ is both a lychrel number and a palindrome. But $k(a)$ is obtained from a by the iterative application of proposition 3.1 to a. By so doing, $a_2...a_{l(a)-1}$ always remains the same for all the iterative numbers obtained, including $k(a)$. Hence, $k(a)_2...k(a)_{l(a)-1} = a_2...a_{l(a)-1}$ and since $k(a)$ is a palindrome, $k(a)_2...k(a)_{l(a)-1}$ is also a palindrome and then $a_2...a_{l(a)-1}$ is a palindrome too. Moreover, when proposition 3.1 is applied to a number a to obtain a number m, one always have $a_1 + a_{l(a)} = m_1 + m_{l(m)}$. We then have $a_1+a_{l(a)} = k(a)_1+k(a)_{l(k(a))}$ and since $k(a)$ is a palindrome, $k(a)_1 = k(a)_{l(k(a))}$ and then $k(a)_1 + k(a)_{l(k(a))}$ is even. It then comes that $\{a_1, a_{l(a)}\}$ is a couple of digits with the same parity.

Conversely, we suppose that $a = a_1a_2...a_{l(a)-1}a_{l(a)}$ is a Lychrel number with $a_2...a_{l(a)-1}$ been a palindrome and $\{a_1, a_{l(a)}\}$ a couple of digits with the same parity. We set $n = \max(a_1, a_{l(a)}) - \frac{a_1+a_{l(a)}}{2}$. Then after iteratively applying proposition 3.1 to a n times, the number $k(a) = a_1a_2...a_{l(a)-1}a_{l(a)} + \frac{a_{l(a)}-a_1}{\mathsf{a}_{l(a)} - a_1}.n.(10^{l(a)-1} - 1)$ is obtained and is a palindrome. Since $a = a_1a_2...a_{l(a)-1}a_{l(a)}$ is a Lychrel number, we conclude that $k(a)$ is a Palindrome-Lychrel number. We can now present the following result:

Proposition 6.1 *Let $a = a_1a_2...a_{l(a)-1}a_{l(a)}$ be a Lychrel number in base b such that $3 \leq b \leq 10$. Then the following assertions are equivalent:*

1. *A Palindrome-Lychrel number can be obtained from a;*
2. *$a_2...a_{l(a)-1}$ is a palindrome and $\{a_1, a_{l(a)}\}$ a couple of digits with the same parity.*

Remark 6.1 *The above result also hold for Lychrel candidates and explain in base* 10 *why Lychrel candidates* 1997, 7779 *or* 18887 *generate Palindrome-Lychrel candidates in their respective equivalence classes.*

Remark 6.2 *It is important to note that even if $a = a_1a_2...a_{l(a)-1}a_{l(a)}$ is not a Lychrel number, one can still obtain a palindrome if $a_2...a_{l(a)-1}$ is a palindrome and $\{a_1, a_{l(a)}\}$ a couple of digits with the same parity. For example, when we consider the natural number* 26568*, we can see that the digits* 2 *and* 8 *have the same parity and* 656 *is a palindrome. Hence setting $n = \max(2, 8) - \frac{2+8}{2} = 3$, we can conclude that after iteratively applying proposition 3.1 to* 26568 3 *times, we will reach a palindrome as follows:* $26568 + 9999 = 36567$, $36567 + 9999 = 46566$ *and* $46566 + 9999 = 56565$ *which is a palindrome.*

7 Conclusion

In this paper, we have defined some tools to explain how to formally obtain an iteration of a natural number in base b such that $3 \leq b \leq 10$. That permits us to provide some properties of constructing new Lychrel numbers or Lychrel candidates from others without using a computer programm. Also, it enables us to present a condition to obtain Palindrome-Lychrel numbers or candidates. Our next steps is to characterize Lychrel numbers from the algebraic point of view and not from the iteration point of view. Moreover, despite the present work, we are still to prove in base 10 that some of the following numbers are Lychrels: 196, 879, 1997, 7059. Hence, we still have to continue looking for more properties of Lychrel numbers and Palindrome-Lychrel numbers which can make their manipulation more easy.

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Figures

196	689
295	788
394	**887**
493	986
592	879
691	978
790	

Figure 1

Equivalence classes of 196 and 887

1495	1585	**1675**	1765	1855	1945
2494	2584	2674	2764	2854	2944
3493	3583	3673	3763	3853	3943
4492	4582	4672	4762	4852	4942
5491	5581	5671	5761	5851	5941
6490	6580	6670	6760	6850	6940

Figure 2

Equivalence class of 1675

1497	1587	1677	1767	**1857**	1947
2496	2586	2676	2766	2856	2946
3495	3585	3675	3765	3855	3945
4494	4584	4674	4764	4854	4944
5493	5583	5673	5763	5853	5943
6492	6582	6672	6762	6852	6942
7491	7581	7671	7761	7851	7941
8490	8580	8670	8760	8850	8940

Figure 3

Equivalence class of 1857

1997
2996
3995
4994
5993
6992
7991
8990

Figure 4

Equivalence class of 1997

4079	4169	4259	4349	4439	4529	4619	4709	4799	4889	4979
5078	5168	5258	5348	5438	5528	5618	5708	5798	5888	5978
6077	6167	6257	6347	6437	6527	6617	6707	6797	6887	6977
7076	7166	7256	7346	**7436**	7526	7616	7706	7796	7886	7976
8075	8165	8255	8345	8435	8525	8615	8705	8795	8885	8975
9074	9164	9254	9344	9434	9524	9614	9704	9794	9884	9974

Figure 5

Equivalence classes related of 7436

8079	8169	8259	8349	8439	8529	8619	8709	8799	8889	8979
9078	9168	9258	9348	**9438**	9528	9618	9708	9798	9888	9978

Figure 6

Equivalence classes related of 9438

8089	8179	8269	8359	8449	8539	8629	8719	8809	8899	8989
9088	9178	9268	9358	9448	9538	9628	9718	9808	9898	**9988**

Figure 7

Equivalence classes related of 9988

7059	7149	7239	7329	7419	7509	7599	7689	7779	7869	7959
8058	8148	8238	8328	8418	8508	8598	8688	**8778**	8868	8958
9057	9147	9237	9327	9417	9507	9597	9687	9777	9867	9957

Figure 8

Equivalence classes related of 7059

12793	**13783**	14773	15763	16753	17743	18733	19723
22792	23782	24772	25762	26752	27742	28732	29722
32791	33781	34771	35761	36751	37741	38731	39721
42790	43780	44770	45760	46750	47740	48730	49720

Figure 9

Equivalence class of 13783

16797	**17787**	18777	19767
26796	27786	28776	29766
36795	37785	38775	39765
46794	47784	48774	49764
56793	57783	58773	59763
66792	67782	68772	69762
76791	77781	78771	79761
86790	87780	88770	89760

Figure 10

Equivalence class of 17787

17897	**18887**	19877
27896	28886	29876
37895	38885	39875
47894	48884	49874
57893	58883	59873
67892	68882	69872
77891	78881	79871
87890	88880	89870

Figure 11

Equivalence class of 18887

13596	14586	15576	**16566**	17556	18546	19536
23595	24585	25575	26565	27555	28545	29535
33594	34584	35574	36564	37554	38544	39534
43593	44583	45573	46563	47553	48543	49533
53592	54582	55572	56562	57552	58542	59532
63591	64581	65571	66561	67551	68541	69531
73590	74580	75570	76560	77550	78540	79530

Figure 12

Equivalence class of 16566

```
i=0
o=0
num=4994
rev_num_str=str(num)[::-1]
if type(num)==float:
    reverse_num=float(rev_num_str)
elif type(num)==int:
    reverse_num=int(rev_num_str)
nu=num+reverse_num
while (o!=nu) and (i!=250000):
    i=i+1
    nu=nu+o
    rev_num_str=str(nu)[::-1]
    if type(nu)==float:
        reverse_num=float(rev_num_str)
    elif type(nu)==int:
        reverse_num=int(rev_num_str)
    o=reverse_num
print(nu)
print(len(str(nu)))
print(i)
```

Figure 13

A Code for computing Lychrel Candidates

Number	8 778	4 994	19 723	89 760	58 883	43 593
β (Threshold)	100 000	250 000	250 000	250 000	250 000	250 000
Time took by the program to reach the Threshold iteration	1h	24h	24h	24h	24h	24h
First ten digits of the βth iteration	1772866159	1765322217	9052771407	9954280876	1574808897	9422139596
Ten last digits of the βth iteration	8951558287	1722223566	6932871508	6889824688	8688917385	6960412238
Number of digits of the βth iteration	41 436	103 464	103 566	103 622	103 465	103 706

Figure 14

Some results with thresholds of 100000 and 250000

Number	8 778	4 994	19 723	89 760	58 883	43 593
β (Threshold)	530 345	531 067	530 914	530 520	531 051	530 404
Time took by the program to reach the Threshold iteration	8 days	8 days	8 days	8 days	8 days	8 days
First ten digits of the βth iteration	1026066438	9128795297	9432996208	8539096138	6617736506	1054105771
Ten last digits of the βth iteration	7844660520	7034988208	7917003338	9406009447	6046377166	3177502549
Number of digits of the βth iteration	220 013	219 799	219 927	219 969	219 793	220 037

Figure 15

Some results with thresholds greater than 500000

Supplementary Files

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